

Final V44 (ALPO) BIT 2-3 OOP 3/13/24 ~~for~~

Part A: Variable Identification Table

Variable Name	Data Type	Variable Type	Stored In
owner	String	Instance Variable	Account object
balance	double	Instance Variable	Account Object
acc1	Account	local reference variable	main() method stack
acc2	Account	local reference variable	main() method stack
acc3	Account	local reference variable	main() method stack

Part B: Explanation

Statement	What happens?	Memory Impact
Account acc1 = new Account("Carlos", 5000);	Creates a new Account object with owner Carlos and balance 5000	Memory allocated in heap for object, acc1 stores reference
Account acc2 = new Account("Diana", 7000);	Creates another Account object	Another object stored in heap
Account acc3 = acc1;	acc3 now refers to the same object as acc1	No new object created, two references point to same memory
acc2.balance = 8000;	Changes Diana's balance to 8000	Modifies values inside acc2 object
acc3.owner = "Miguel";	Changes owner of the object acc1 and acc3 refer to	updates the same shared object in heap

Part C: Output

Miguel 5000.0
Diana 8000.0
Miguel 8000.0